

PAPER

Improved APF-based real-time path planning and control of underactuated AUVs in complex marine environments

To cite this article: Longjin Wang *et al* 2025 *Meas. Sci. Technol.* **36** 046209

View the [article online](#) for updates and enhancements.

You may also like

- [Trajectory tracking control of autonomous underwater vehicles based on a predefined time control strategy](#)
Yang Liu, Longjin Wang, Shun An *et al.*
- [A 3D adaptive spiral coverage path planning algorithm of autonomous underwater vehicle for enhanced edge-corner coverage](#)
Zhaoye Chen, Jiaxin Gao, Jiahui Ma *et al.*
- [Multiple-AUV formation control with a collision avoidance algorithm based on a virtual structure and an improved artificial potential field method](#)
Jian Wang, Zhen Hu and Hua Zhang

Precision or Throughput? Why Choose?

Next-generation photonic manufacturing requires nanometer precision, scalable automation, and the flexibility to adapt.

SmarAct's motion and alignment solutions combine high-dynamic positioning, automated optical alignment, and integrated metrology for demanding photonic assembly and testing applications. Flexible system architectures support scalable integration processes across a broad range of optical technologies and advanced manufacturing environments.



- Nanometer Precision
- Automated Alignment
- Integrated Metrology
- Modular Architecture
- Scalable Manufacturing

Enable Scalable Optical Assembly

smaract.com



Improved APF-based real-time path planning and control of underactuated AUVs in complex marine environments

Longjin Wang, Zhiyang Wang, Yang Liu, Shun An* , Fang Deng and Zhimin Fan

College of Electrical and Mechanical Engineering, Qingdao University of Science and Technology, Qingdao 266061, People's Republic of China

E-mail: anshun111@126.com

Received 6 December 2024, revised 10 March 2025

Accepted for publication 11 March 2025

Published 20 March 2025



Abstract

This paper proposes an integrated path planning and tracking method for autonomous underwater vehicles (AUVs) to maintain a path without collision in complex marine environments. For the path planning approach, an improved artificial potential field algorithm is proposed by introducing and combining a logarithmic function, a distance compensation term, a virtual obstacle construction method, and a velocity repulsive potential field function, which effectively solves the problem of the goal non-reachable with obstacle nearby, local minimum and dynamic obstacles avoidance. For the tracking control method, an adaptive fixed-time integral sliding mode controller is presented to track the planned path for collision avoidance maneuvers. Stability analysis proves that the proposed control strategy can make the tracking errors of the AUV converge to arbitrarily small neighborhood of the origin within a fixed-time independent of the initial conditions. The results of the simulation show that the proposed approach is capable of effectively achieving real-time path planning and tracking control of AUVs in marine environments with complex obstacle.

Keywords: autonomous underwater vehicles, fixed-time control, path planning, artificial potential field, dynamic obstacles avoidance

1. Introduction

Autonomous underwater vehicle (AUV) is a kind of unmanned vehicles that can autonomously perform certain tasks in complex marine environment [1, 2]. With the continuous development of underwater measurement technology, AUV's ability to operate in complex underwater environments has been improved. Intelligent navigation of AUVs is increasingly attracting attention [3–5]. With the increasing complexity of application scenarios, the randomness and instability of obstacles will directly affect the safety and efficiency of navigation, so solving the problem of real-time path planning

for AUVs in dynamic and unknown environments is of great research significance in related industries [4, 6, 7].

At present, researchers have conducted some studies on the real-time path planning for AUVs [8, 9]. In [10], a closed-loop rapid exploration random tree algorithm is proposed for path planning of feasible paths of AUVs from initial position and velocity to target position and velocity in 3D space. In [11], an improved dynamic bionic neural network is proposed for path planning of AUVs, which solves the complex computational problems in large 3D environments. However, the above researchs require a large amount of computational space and cannot quickly deal with unknown dynamic obstacles.

Artificial potential field (APF) algorithm is a real-time path planning method proposed in [12]. The vehicles can approach the target point and avoid collision with obstacles

* Author to whom any correspondence should be addressed.

by establishing an APF. The APF algorithm has been widely used because of its advantages of simple mathematical calculation and convenient real-time observation. However, the APF algorithm have some limitations such as the goals non-reachable with obstacle nearby (GNRON), local minimum, and dynamic obstacles avoidance problems [13]. Some researchers had solved these problems by combining APF algorithms with other algorithms. In [14], an A*+APF based path planning algorithm for is proposed AUVs to achieve fast and effective obstacle avoidance, which solves the problem of slow operation of the A* algorithm in complex multi-obstacle environments. Nevertheless, among these studies, the APF algorithm was only combined with other global intelligence algorithms to tackle the path planning problem of AUVs, but it does not improve the APF algorithm to solve the limitations.

To address these inherent drawbacks of the APF algorithm, researchers have made some contributions. In [15], a improved APF algorithm is proposed by redefining the attractive potential field function as a logarithmic function, which avoids the local oscillations of the three-point covariance and semi-enclosed obstacles. In [16], a model predictive APF path planning with considering collision avoidance rules method is proposed, which transforms the problem of motion planning into a nonlinear optimisation problem with multiple constraints, such as manoeuvrability, navigational rules, and navigable channels, and solves the local optimisation problem of the traditional APF algorithm. For the problem of obstacle avoidance in the AUV formation., a cooperative obstacle avoidance algorithm based on improved APF algorithm and consistency agreement is proposed in [17]. An auxiliary potential field is designed that is perpendicular to the motion direction of the AUVs. This solves the problem of the combined force tending to zero at the local minimum of the potential field. The above studies improve the APF algorithm to avoid the local minimum problem. However, these studies have not solved the problem of GNRON and dynamic obstacles avoidance. In [18], an novel APF method based on virtual navigation paths generated by an adaptive particle swarm optimisation algorithm is proposed, which solves the local minimum and GNRON problems in the traditional APF. For the dynamic obstacles avoidance problem. In [19], a virtual potential field detection circle model (VPFDCM) with adjustable radius is proposed for detecting the local minimum potential field value of the obstacle repulsive field in advance, and combined with an improved reinforcement learning algorithm based on long-short-term memory, the radius of the VPFDCM is adjusted, which achieves effective obstacle avoidance for dynamic obstacles. Nevertheless, all of the above studies treat the AUVs as particles, and studies their motions under fixed time steps, without considering practical applications. In complex environments, real-time guidance will frequently change the heading angle of the AUV, while there exist the effects of wave, current and other perturbations, which requires fast and accurate tracking control to avoid various obstacles to reach the target position.

Commonly used control methods for AUV tracking control include sliding mode control [20], backstepping control

[21, 22], neural network control [23, 24], and fuzzy control [25]. Sliding mode control is considered one of the most effective methods for dealing with nonlinear systems with uncertainties and external disturbances due to its strong robustness [26]. In [27], a disturbance observer based nonsingular fast terminal sliding mode control scheme is proposed for the trajectory tracking control of AUVs, which improves the convergence speed and robustness to against external disturbances. In [28], a new robust controller for underactuated AUV horizontal plane path tracking based on backstepping and sliding mode control with a novel sliding surface is proposed to address the path tracking control in the presence of modelling errors, fast time-varying external disturbances, and the absence of velocity measurement data. In [29], a control scheme with a finite-time performance function and the fast terminal sliding surface is proposed for tacking control of AUVs. However, above these control algorithms can only achieve asymptotic or finite time stability, and the convergence time depends on the initial state of the system, which implies that the tracking errors of AUVs take a long time to converge to the equilibrium point.

In summary, the research on real-time path planning algorithms and control methods for AUVs in complex dynamic marine environments is currently inadequate. This paper proposes an integrated path planning and control scheme for AUVs to plan a safe path in complex marine environments. The overall technical solution is shown in the figure 1. The main contributions of this paper are as follows:

- By introducing and combining a logarithmic function, a distance compensation term, a virtual obstacle construction method, and a velocity repulsive potential field function, an improved APF algorithm is proposed for AUVs to achieve real-time path planning in the environments with complicated obstacles , which effectively overcomes drawbacks of GNRON, local minimum and dynamic obstacles avoidance inherent in the traditional APF algorithm.
- An adaptive fixed-time integral sliding mode control (AFISMC) scheme is presented for tracking control of the AUV. Under the proposed controller, the tracking errors can stabilize to small region of the origin within a fixed-time independent of the initial conditions.

An outline of this paper is summarized as follows. In section 2, the mathematical model of the AUV is given. Section 3 proposes a improved APF algorithm, which effectively solves problems of the GNRON, local minimum, and dynamic obstacle. Section 4 designs the adaptive fixed-time integral sliding mode tracking controller. Section 5 establishes the system simulations and verifies the validity of the algorithm. Section 6 summarizes the research results.

2. The mathematical model for the AUV

This paper explores the problem of real-time local path planning in complex environments. Shallow water environments are of particular interest due to the increased presence of

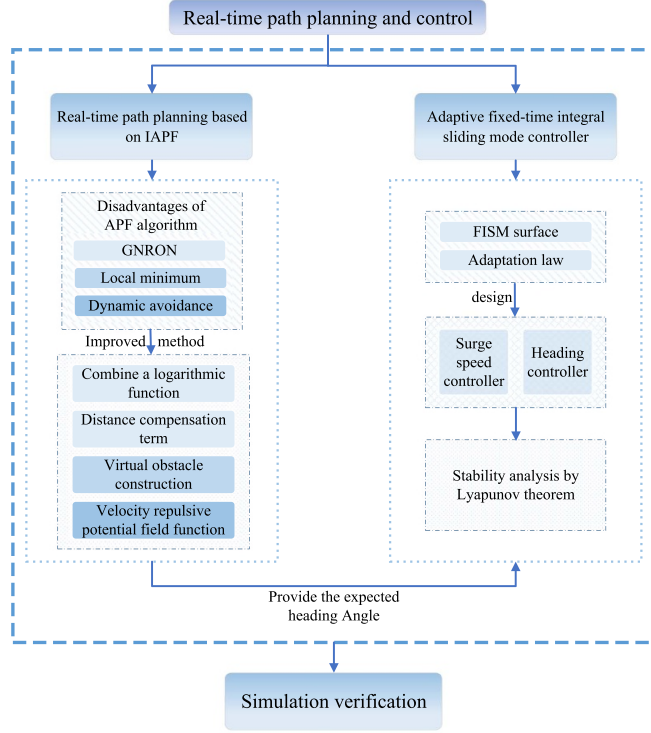


Figure 1. A flowchart of the overall technical solution.

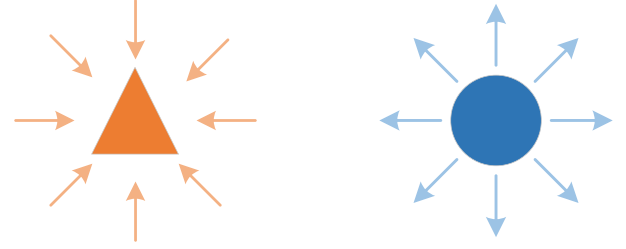
obstacles and disturbances compared to deep sea environments. As most shallow water areas have limited depth variations, for the purpose of analysis, the motion of the AUV is typically described in both the inertial frame and the bodyfixed frame. By ignoring the roll, pitch and heave motions, the kinematic and dynamic model of the AUV in the horizontal plane can be expressed as [30]

$$\begin{cases} \dot{x} = u \cos(\psi) - v \sin(\psi) \\ \dot{y} = u \sin(\psi) + v \cos(\psi) \\ \dot{\psi} = r \end{cases} \quad (1)$$

$$\begin{cases} m_{11}\dot{u} = m_{22}vr - d_{11}u + d_{uu}|u|u + \tau_u + d_u \\ m_{22}\dot{v} = -m_{11}ur - d_{22}v + d_{vv}|v|v + d_v \\ m_{33}\dot{r} = (m_{11} - m_{22})uv - d_{33}r + d_{rr}|r|r + \tau_r + d_r \end{cases} \quad (2)$$

where (x, y) and ψ express the position and heading angle of the AUV in the earth-fixed coordinate system, u and v express the linear velocities of the AUV in surge and sway, r indicate the angular velocity of the AUV in yaw, τ_u and τ_r indicate the control inputs generated by the propeller and rudder, and d_u , d_v , d_r denote the unexplored environmental disturbances, m_{ii} ($i = 1, 2, 3$) are considered to be the sum of the inertial and added mass terms, d_{11} , d_{22} and d_{33} , d_{uu} , d_{vv} and d_{rr} denote the linear and nonlinear hydrodynamic parameters respectively.

Assumption 1. The disturbances d_u, d_v, d_r are assumed to unknown time-varying but bounded such that there are unknown positive constants ρ_u, ρ_v, ρ_r satisfying $|d_u| \leq \rho_u, |d_v| \leq \rho_v, |d_r| \leq \rho_r$.



(a) Attractive potential field (b) Repulsive potential field

Figure 2. Diagram of the potential field.

3. Real-time path planning based on improved APF algorithm

3.1. Traditional APF algorithm

The APF algorithm is a virtual force algorithm that was proposed in [31]. It is a classical path planning algorithm that is widely used in local path planning. This section describes a method for simulating a potential field in the AUV working scenario, similar to an electromagnetic field. As shown in figures 2(a) and (b), targets and obstacles generate potential fields. The AUV experiences attractive force from the target and repulsive force from the obstacles in the artificially created potential field, causing it to move in the direction of the combined force.

The AUV is constantly affected by the attractive force. The attractive force increases with the distance between the AUV and the target increasing. The attractive potential field function U_{att} is formulated as follows:

$$U_{att}(X) = \frac{1}{2}k_{att}d^2(X, X_t) \quad (3)$$

where X denote current position coordinate of AUV, and X_t denote target position coordinates, $d(X, X_t) = \|X_t - X\|$ denote the distance between the AUV and target, and k_{att} is the attractive potential field parameter. The attractive force $F_{att}(X)$ is the negative gradient of $U_{att}(X)$

$$F_{att}(X) = -\nabla U_{att}(X) = k_{att}d(X, X_t) \vec{i}(X, X_t) \quad (4)$$

where $\vec{i}(X, X_t)$ is the unit vector along the AUV pointing in the direction of the target, indicating the direction of attraction.

The position coordinates of obstacles is denoted as X_o . The obstacle generates a repulsive potential field, which is a local potential field with an influence range determined by the radius of obstacle. When the AUV is within the scope of influence of obstacles, it is subject to a repulsive force that increases as the AUV approaches obstacles. The repulsive potential field function $U_{rep}(X)$ is denoted as:

$$U_{rep}(X) = \begin{cases} \frac{1}{2}k_{rep} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_o} \right)^2, & d(X, X_o) \leq d_o \\ 0, & d(X, X_o) > d_o \end{cases} \quad (5)$$

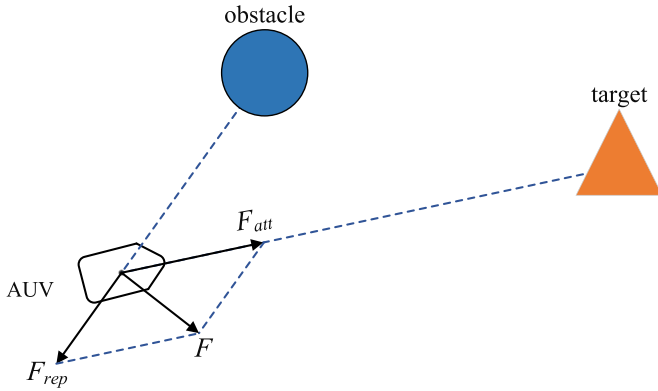


Figure 3. Potential field analysis of the AUV in artificial potential field.

where $d(X, X_o) = \|X_o - X\|$ denote the distance between the AUV and the obstacle, k_{rep} is the repulsive potential field parameter. d_0 indicate the radius of the area of influence of the obstacle. The repulsive force $F_{rep}(X)$ is a negative gradient of $U_{rep}(X)$, which denoted as:

$$F_{rep}(X) = -\nabla U_{rep}(X) = \begin{cases} k_{rep} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right)^2 \frac{\nabla d(X, X_o)}{d^2(X, X_o)} \vec{i}(X, X_o), & d(X, X_o) \leq d_0 \\ 0, & d(X, X_o) > d_0 \end{cases} \quad (6)$$

where $\vec{i}(X, X_o)$ is a unit vector indicates the direction of the repulsive force, which pointing towards AUVs along the centre of obstacles.

The AUV is subjected to a combined attractive and repulsive potential field during its motion, as illustrated in figure 3. The combined potential field function $U(X)$ of the AUV is expressed as:

$$U(X) = U_{att}(X) + U_{rep}(X). \quad (7)$$

According to the principle of superposition the combined force $F(X)$ is:

$$F(X) = -\nabla U_{att}(X) - \nabla U_{rep}(X) = F_{att}(X) + F_{rep}(X). \quad (8)$$

The conventional APF algorithm can address local obstacle avoidance in simple scenes. It has the advantage of requiring less computation and being better equipped to handle unknown obstacle threats compared to other classical obstacle avoidance algorithms. However, there are still some limitations.

The repulsive force F_{rep} , which increases as the AUV approaches the obstacle, is in opposition to the attractive force F_{att} , which decreases as the AUV approaches the target. This results in the problem of unreachable target points when the obstacle is near the target. In complex obstacle environments, the AUV may experience a situation where the attractive and repulsive forces balance each other out, or where the forces are in the same direction, resulting in the AUV being locally stagnant or oscillating (Hereafter, we will refer to the

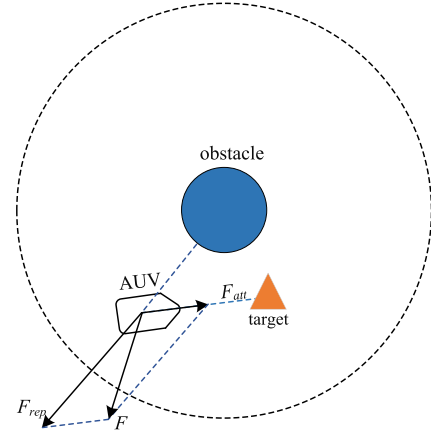


Figure 4. Force analysis of GNRON problem.

local minimum problem). The conventional APF algorithm fail to take into account the effect of velocity, despite its ability to solve dynamic obstacles at lower speeds. However, it is unable to effectively avoid dynamic obstacles at higher speeds.

Therefore, in order to solve the target unreachable, local minimum, and dynamic obstacle problems of the traditional APF algorithm, this paper proposes an improved APF algorithm, which improves the adaptability of the algorithm to the complex navigation environment.

3.2. Improved APF algorithm

3.2.1. GNRON problem. When the AUV is near the obstacle, according to (4) and (6), the distance $d(X, X_o)$ between the AUV and the obstacle tends to zero, and its inverse $\frac{1}{d(X, X_o)}$ tends to infinity. Consequently, as the AUV approaches the target, the attractive force decreases, while the repulsive force increases. Figure 4 illustrates the GNRON problem, where the dashed line indicates the sphere of influence of the repulsive potential field. The repulsive force on the AUV is much greater than the attractive force, resulting in a combined force that always points away from the target point. As a result, it is impossible to reach the target point.

To solve the above problem, two enhancements have been made to the potential field function. Function (4) shows that the attractive potential field function of the traditional APF algorithm is a proportional function. Consequently, the attractive force increases monotonically in direct proportion to the distance between the AUV and the target. When the distance is considerable, the attractive forces generated can be excessive, which may result in collisions with obstacles. As the AUV approaches the target, the attractive force decreases gradually. To address these issues, the attractive potential field function of the target has been improved as follows:

$$U_{att}(X) = \begin{cases} k_{att1} \left[d(X, X_t) \ln \left(\frac{d(X, X_t)}{d(X, X_t) + 1} \right) - \ln(d(X, X_t) + 1) \right], & d(X, X_t) \leq d_t \\ k_{att2} [(d(X, X_t) + 1) \ln(d(X, X_t) + 1) - (d(X, X_t) + 1)], & d(X, X_t) > d_t \end{cases} \quad (9)$$

where k_{att1} and k_{att2} are the attractive potential field parameters, $d(x, x_t)$ is the euclidean distance between the AUV and the target, and d_t is the distance parameter between the AUV and the target.

Considering that at locations where the AUV is far from the target, the attractive force should increase as the distance between the two increases to avoid the risk of colliding with obstacles, while at locations where the AUV is close to the target, the attractive force should increase as the distance decreases to avoid the GNRRN problem.

Therefore, attractive potential field function is designed as the segmented function and a logarithmic function is introduced, and the improved attractive function is the negative gradient of the attractive potential field function, denoted as :

$$F_{att}(X) = -\nabla U_{att}(X) = \begin{cases} k_{att1} \left| \ln \left(\frac{d(X, X_t)}{d(X, X_t) + 1} \right) \right| \vec{i}(X, X_t), & d(X, X_t) \leq d_t \\ k_{att2} \left| \ln(d(X, X_t) + 1) \right| \vec{i}(X, X_t), & d(X, X_t) > d_t. \end{cases} \quad (10)$$

From (10) it can be seen that at $d(X, X_t) \leq d_t$ the attractive function is a monotonically decreasing function and the attractive force increases as the distance from the target decreases, whereas at $d(X, X_t) > d_t$, the attractive function is a monotonically increasing function, and considering the characteristics of the logarithmic function, the problem of collision with obstacles due to excessive attractive force is solved as the attractive force grows smoothly.

On the other hand, the increased repulsive force is balanced by adding a compensation term pointing in the direction of the target, which gradually decreases as the AUV approaches the target to ensure that the AUV can reach the target point. The enhanced repulsive potential field function is expressed as:

$$U_{rep}(X) = \begin{cases} \frac{1}{2} k_{rep} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right)^2 d^n(X, X_t), & d(X, X_o) \leq d_0 \\ 0, & d(X, X_t) > d_0. \end{cases} \quad (11)$$

The distance term $d^n(X, X_t)$ from the AUV to the target is added to (5), where n is any real number greater than zero. The enhanced repulsive force function is the negative gradient of the repulsive potential field function:

$$F_{rep}(X) = -\nabla U_{rep}(X) = \begin{cases} F_{rep1}(X) + F_{rep2}(X), & d(X, X_o) \leq d_0 \\ 0, & d(X, X_t) > d_0 \end{cases} \quad (12)$$

$$F_{rep1}(X) = k_{rep1} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right) \frac{d^n(X, X_t)}{d^2(X, X_o)} \vec{i}(X_o, X) \\ F_{rep2}(X) = k_{rep1} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right) d^{n-1}(X, X_t) \vec{i}(X, X_t). \quad (13)$$

The force analysis under the GNRRN problem is shown in figure 4, where the attractive force F_{att} is increased compared

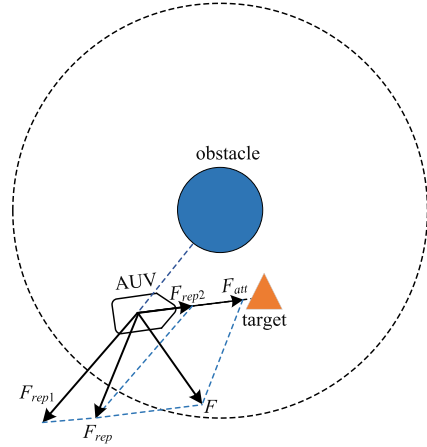


Figure 5. Force analysis under improved APF algorithm.

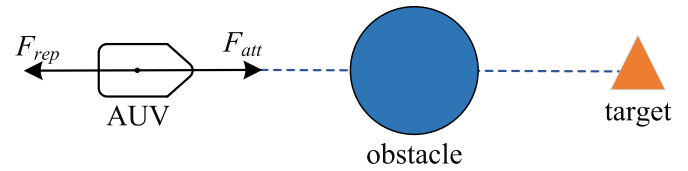


Figure 6. AUV, target and obstacle collinear diagram.

to the attractive force under the traditional APF algorithm (shown in figure 5). This is due to the addition of the repulsive force component F_{rep1} to the target orientated underwater vehicle, which brings the combined force closer to the target and avoids the GNRRN problem.

3.2.2. Local minimum problem. As the AUV moves towards the target, it may pass through potential field minima en route, points at which attractive and repulsive forces cancel each other out. As shown in figure 6, the attractive force F_{att} gradually decreases and the repulsive force F_{rep} gradually increases, and there exists a position $F_{att} = F_{rep}$ where the combined force applied is zero and the AUV stagnates or locally oscillates. The local minimum problem in the case of multiple obstacles is shown in figure 7, where the attractive force F_{att} and repulsive force F_{rep} cancel each other out and the same phenomenon of AUV stagnation or local oscillation occurs.

To solve the local minimum problem, this paper proposes a virtual obstacle circle construction method. When the AUV navigates to the local minimum area, the virtual obstacle circle is constructed so that its centre is offset from the centre of the original obstacle, as shown in figure 8, and it completely contains the original obstacle to ensure safe navigation. The centre O' of the virtual obstacle circle and the radius R' are denoted as:

$$O'(x', y') = (x, y) + o_i \vec{o}_i \quad (14)$$

$$R' = \sqrt{2} R \cdot o_i \quad (15)$$

where $O'(x', y')$ is the centre coordinate of the virtual obstacle circle, (x, y) is the centre coordinate of the original obstacle,

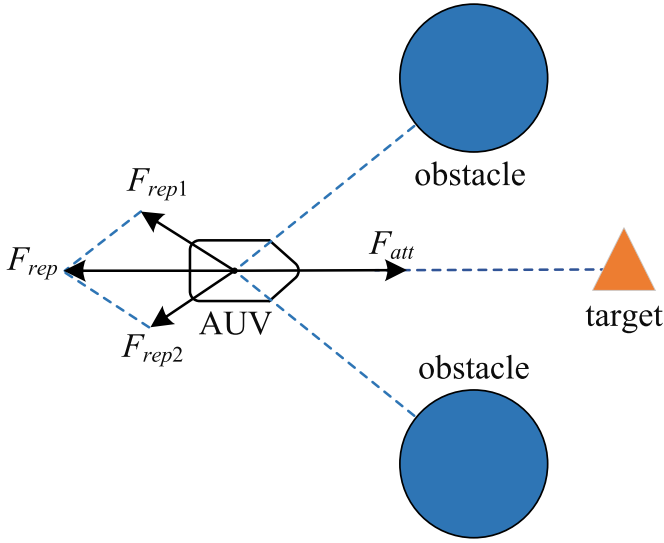


Figure 7. Local minimum problem diagram of potential field under multiple obstacles.

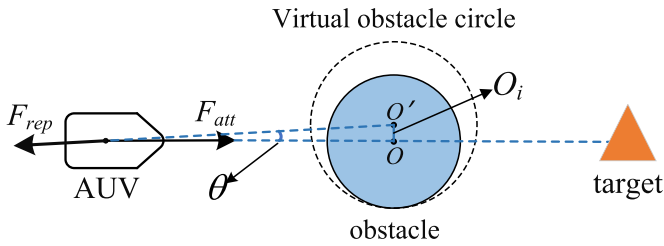


Figure 8. Diagram of virtual obstacle circle.

R' is the radius of the virtual obstacle circle, R is the radius of the original obstacle and o_i is the eccentricity parameter of the virtual obstacle circle. $\vec{o}_i$ represents the unit vector in the vertical direction of the target along the AUV.

As shown in figure 8, there is an angle θ between the AUV and the line connecting the target and the centre of the virtual obstacle circle. Although the value of angle θ is small when the AUV is far away from the obstacle, the AUV will gradually move away from the obstacle under the influence of the repulsive potential field generated by the virtual obstacle circle, so that the angle θ will continue to increase and the obstacle will be avoided.

For a situation in which there are multiple obstacles, a virtual obstacle circle is constructed for the obstacle closest to the AUV. Similarly, the AUV will not stagnate or oscillate when the local potential field minimum are broken, as shown in figure 7.

3.2.3. Dynamic obstacles avoidance problems. The traditional APF algorithm has less influence when the obstacle is moving at a low speed, but when the obstacle is moving faster, the traditional APF does not take into consideration the effect of velocities of obstacles, and the following two situations will occur:

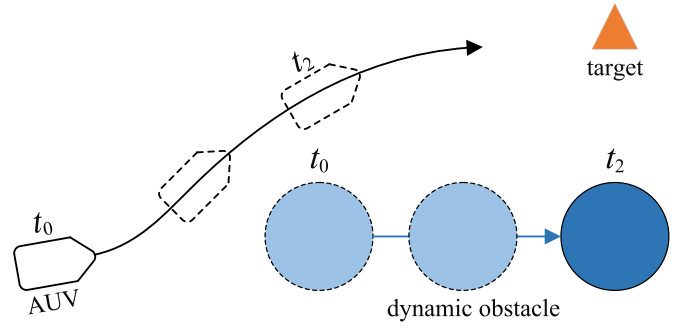


Figure 9. Diagram of AUV deviation from dynamic obstacles.

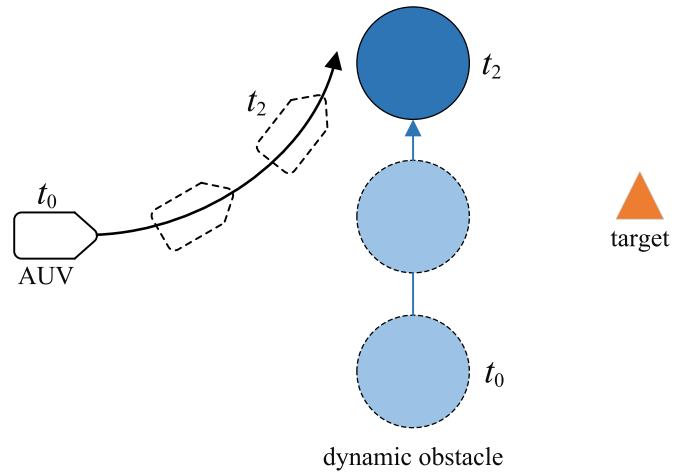


Figure 10. Diagram of AUV approaching obstacles.

- At t_0 in figure 9, the AUV is in range of a dynamic obstacle and begins to avoid the obstacle. In fact, at this time, the dynamic obstacle is moving away from the AUV at high speed, and the AUV is about to leave the obstacle's influence range, and there is no risk of collision even if the AUV does not change its heading, but only moves towards the target point under the effect of gravity. This results in a longer path and reduces navigation efficiency.
- At t_0 in figure 10, The AUV calculates the obstacle avoidance heading based on the traditional APF theory. Due to the fast speed of dynamic obstacles, the heading cannot be dynamically adjusted in time, resulting in a high risk of collision between the AUV and the obstacle.

To address the above problems, this paper proposes the relative direction comparison method to improve the traditional APF algorithm.

For the purpose of problem analysis, the AUV navigation posture is constructed as shown in figure 11, where V_a is the speed of the AUV, V_o is the velocity of the dynamic obstacle, V_{ao} is the relative velocity of the AUV to the obstacle, l_1 is the line connecting the AUV and the obstacle, l_2 is the normal to l_1 the line, α is the angle between the direction of the relative velocity and the direction of the relative position, β is the angle between the tangent line l_3 to the maximum safe range of the

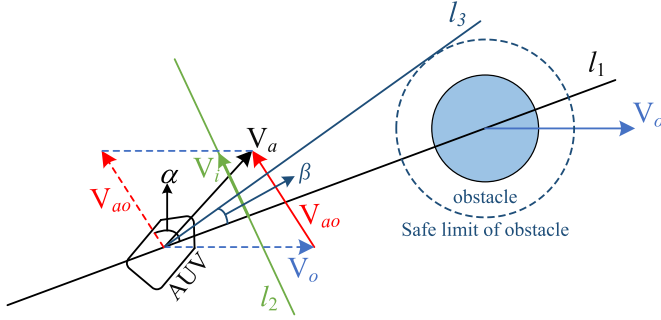


Figure 11. AUV deviation obstacle analysis under improved artificial potential field.

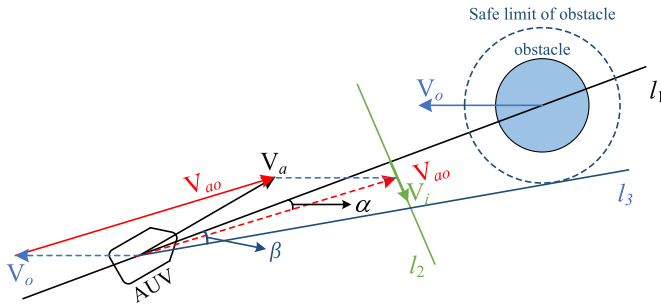


Figure 12. AUV approaching obstacle analysis diagram under improved artificial potential field.

AUV and the obstacle and the line l_1 , and V_i is the unit vector of the component of V_{ao} in the L_2 direction.

The safety range limit of the obstacle is set according to the minimum safe encounter distance of the AUV. The radius of the general safety range is the radius of the obstacle plus 1.5 times the minimum safe encounter distance. The minimum encounter distance is determined according to international collision avoidance regulations.

As shown in figure 12, when $\alpha < \beta$, in the collision hazard analysis, it can be understood that the obstacle remains stationary, the AUV moves with V_{ao} , the angle between the relative velocity direction and the relative position direction is small, the AUV moves towards the obstacle within the safe collision range, and there is a collision risk between the AUV and the obstacle. Therefore, the potential field function $U_{rep}(X)$ is further improved on the basis of (13), and the magnitude of α and β is used as the judgement condition. The improved repulsive potential field function $U_{rep}(X, V)$ is expressed as:

$$U_{rep}(X, V) = \begin{cases} U(X) + U(V), & d(X, X_o) \leq d_0, \alpha < \beta \\ 0, & \text{else} \end{cases} \quad (16)$$

$$U(X) = \frac{1}{2} k_{rep} \left(\frac{1}{d(X, X_o)} - \frac{1}{d_0} \right)^2 d^n(X, X_i) \quad (17)$$

$$U(V) = k_v \frac{1}{\alpha^2} \frac{V_{ao}}{d(X, X_o)} \quad (17)$$

where k_v is the dynamic obstacle rejection parameter and V_{ao} is the relative velocity of the AUV to the dynamic obstacle.

The repulsive force function is:

$$F_{rep} = F_{rep}(X) + F_{rep}(V) \quad (18)$$

$$F_{rep}(V) = k_v \frac{1}{\alpha^2} \frac{V_{ao}}{d(X, X_o)} V_i. \quad (19)$$

The improved repulsive potential field function avoids blind obstacle avoidance by determining the magnitude of the angle α between the relative velocity direction and the relative position direction, and the angle β between the AUV and the safe area of the obstacle, while increasing the repulsive force away from the relative direction of the obstacle for dynamic obstacles at risk of collision, thereby improving obstacle avoidance in the face of dynamic obstacles.

4. AFISM design

In this section, an AFISM controller is proposed to track the desired surge speed and desired heading angle generates by the improved APF algorithm. Under the action of the proposed control scheme, the tracking errors of the AUV can be stabilized to the region of the origin in fixed time. The proposed real-time path planning and control scheme is shown in the figure 13.

4.1. Fixed-time integral sliding mode surface

To design the AFISM control scheme, the fixed-time integral sliding mode (FISM) surface is introduced as follows

$$\begin{cases} \sigma = z + az_I \\ \dot{z}_I = b_1 \text{sig}^m(z) + b_2 \text{sig}^n(z), \text{ with } z_I(0) = -a^{-1}z(0) \end{cases}, \quad (20)$$

where z is the tracking error, and $a > 0, b_1 > 0, b_2 > 0, 0 < m < 1, n > 1$ are constants.

If the variable σ is maintained at zero, then $z = -az_I$ and (20) can be rewritten as

$$\dot{z}_I = -b_1 a^m \text{sig}^m(z_I) - b_2 a^n \text{sig}^n(z_I). \quad (21)$$

According to lemma 2 in [31], z_I will converge to the origin in fixed time which is determined by

$$t_r = \frac{1}{b_1 a^m (1-m)} + \frac{1}{b_2 a^n (n-1)}. \quad (22)$$

Moreover, the convergence of z is achieved in the same fixed time because $z = -az_I$ on the fixed-time ISM surface $\sigma = 0$.

4.2. Surge speed controller design

The surge speed controller is designed to drive the surge velocity tracking errors converge to small region of the origin. Consider the following first-order surge speed control subsystem

$$m_{11} \dot{u} = m_{22} vr - d_{11} u + d_{uu} |u| u + \tau_u + d_u. \quad (23)$$

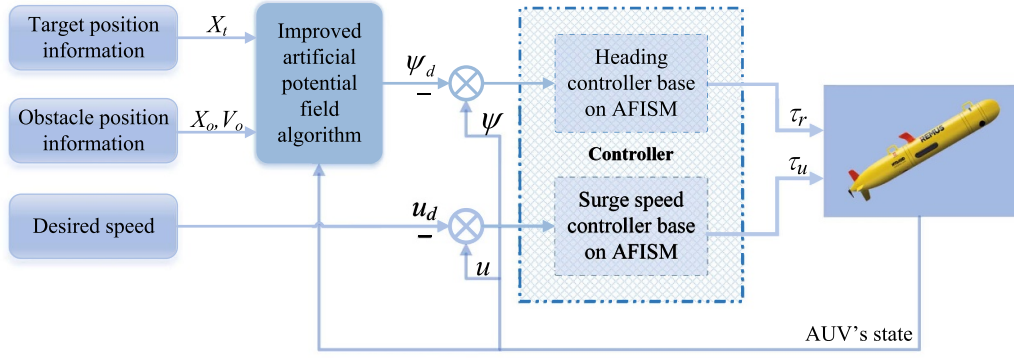


Figure 13. Proposed real-time path planning and control scheme.

Define the surge speed tracking error $z_u = u - u_d$, where u_d is the desired speed of the AUV. According to the tracking error z_u , the dynamic FISM surface in the surge motion direction can be designed as

$$\begin{cases} \sigma_u = m_{11}z_u + a_u z_{I,u} \\ \dot{z}_{I,u} = b_{1u} \text{sig}^m(z_u) + b_{2u} \text{sig}^n(z_u), \text{ with } z_{I,u}(0) = 0 \end{cases}, \quad (24)$$

where $a_u, b_{1u}, b_{2u} > 0$ are design parameters.

Deriving (24) with respect to time and substituting it into (23) yields

$$\dot{\sigma}_u = m_{22}vr - d_{11}u + d_{uu}|u|u + \tau_u + d_u - m_{11}\dot{u}_d + a_u \dot{z}_{I,u}. \quad (25)$$

To ensure that the surge velocity tracking error z_u of AUV converges within a fixed-time, an AFISM surge velocity controller τ_u based on a dynamic FISM surface σ_u can be designed as

$$\begin{aligned} \tau_u = & m_{11}\dot{u}_d - m_{22}vr + d_{11}u - d_{uu}|u|u - a_u \dot{z}_{I,u} \\ & - \hat{\rho}_u \text{sign}(\sigma_u) - k_{1u} \text{sig}^{2p-1}(\sigma_u) - k_{2u} \text{sig}^{2q-1}(\sigma_u), \end{aligned} \quad (26)$$

where $k_{1u} > 0, k_{2u} > 0, 0 < p < 1, q > 1$ are constants. $\hat{\rho}_u$ is the estimation of ρ_u which is determined by the following adaptive law

$$\dot{\hat{\rho}}_u = \lambda_0 |z_u| - \lambda_1 \hat{\rho}_u - \lambda_2 \hat{\rho}_u^{2q-1} \quad (27)$$

where λ_0, λ_1 and λ_2 are positive constants.

4.3. Heading controller design

A heading controller is designed to stabilise the heading tracking error within a fixed-time. Now, consider the following second-order heading control subsystem

$$\dot{\psi} = r \quad (28)$$

$$m_{33}\dot{r} = (m_{11} - m_{22})uv - d_{33}r + d_{rr}|r|r + \tau_r + d_r. \quad (29)$$

Define the heading tracking error $z_\psi = \psi - \psi_d$. Then, the kinematic FISM surface in the yaw motion direction can be designed as

$$\begin{cases} \sigma_\psi = z_\psi + a_\psi z_{I,\psi} \\ \dot{z}_{I,\psi} = b_{1\psi} \text{sig}^m(z_\psi) + b_{2\psi} \text{sig}^n(z_\psi), \text{ with } z_{I,\psi}(0) = 0 \end{cases}, \quad (30)$$

where $a_\psi, b_{1\psi}, b_{2\psi} > 0$ are design parameters.

Differentiating (30) with respect to time, one has

$$\dot{\sigma}_\psi = r + a_\psi \dot{z}_{I,\psi}. \quad (31)$$

To ensure that the heading angle tracking error z_ψ of the AUV converges within a fixed-time, the virtual controller r_c which acts as the reference heading angle velocity command for the heading angle velocity signal r is designed as

$$r_c = -a_\psi \dot{z}_{I,\psi} - l_{1\psi} \text{sig}^{2p-1}(\sigma_\psi) - l_{2\psi} \text{sig}^{2q-1}(z_\psi). \quad (32)$$

According to (32), the virtual controller r_c is too complex, making its differentiation $\dot{r}_c$ difficult to obtain directly through computation. To obtain the differential terms of the virtual controller r_c used in the following control design, a nonlinear first-order filter is introduced

$$\kappa_u \dot{r}_d = -\text{sig}^{2p-1}(\zeta_r) - \text{sig}^{2q-1}(\zeta_r) \quad (33)$$

where $\zeta_r = r_d - r_c$ is the filtered error, r_d is the filtered output, and κ_r is a positive constant. From (33), we can see that the differential term $\dot{r}_d$ can be directly obtained to replace the complex differential term $\dot{r}_c$.

Define the heading angle velocity tracking error $z_r = r - r_d$. Then, the dynamic fixed-time ISM surface in the yaw motion direction can be designed as

$$\begin{cases} \sigma_r = m_{33}z_r + a_r z_{I,r} \\ \dot{z}_{I,r} = b_{1r} \text{sig}^m(z_r) + b_{2r} \text{sig}^n(z_r), \text{ with } z_{I,r}(0) = 0 \end{cases}. \quad (34)$$

Differentiating (34) with respect to time, we can obtain

$$\dot{\sigma}_r = (m_{11} - m_{22})uv - d_{33}r + d_{rr}|r|r + \tau_r + d_r - m_{33}\dot{r}_d + a_r \dot{z}_{I,r}. \quad (35)$$

To guarantee the fixed-time convergence of the heading angle velocity tracking error z_r of the AUV, the adaptive fixed-time ISM heading controller τ_r based on the dynamic fixed-time ISM surface σ_r can be designed as

$$\tau_r = m_{33}\dot{r}_d - (m_{11} - m_{22})uv + d_{33}r - d_{rr}|r|r - a_r\dot{z}_{l,r} - \hat{\rho}_r \text{sign}(\sigma_r) - k_{1r} \text{sig}^{2p-1}(\sigma_r) - k_{2r} \text{sig}^{2q-1}(\sigma_r), \quad (36)$$

where $k_{1r} > 0, k_{2r} > 0, 0 < p < 1, q > 1$ are constants. $\hat{\rho}_r$ is the estimation of ρ_r which is determined by the following adaptive law

$$\dot{\hat{\rho}}_r = \eta_0|z_r| - \eta_1\hat{\rho}_r - \eta_2\hat{\rho}_r^{2q-1} \quad (37)$$

where η_0, η_1 and η_2 are positive constants.

4.4. Stability analysis

The stability of the tracking errors of the AUV can be proved by the following theorem.

Theorem 1. Consider the AUV tracking errors z_u, z_ψ, z_r with the fixed-time ISM surfaces defined in (24), (30) and (34), if the surge speed controller is designed as (26), the virtual controller is designed as (32) and the yaw dynamic controller is designed as (36) with the adaptive laws (27) and (37), then, the tracking errors of the AUV will converge to the neighbourhood of zero in fixed time.

Proof 1. Consider a Lyapunov function as follows:

$$V = \frac{1}{2} \left(\sigma_u^2 + \sigma_\psi^2 + \sigma_r^2 + \zeta_r^2 + \frac{\tilde{\rho}_u^2}{\lambda_0} + \frac{\tilde{\rho}_r^2}{\eta_0} \right). \quad (38)$$

Taking the time derivative of (38) yields

$$\dot{V} = \sigma_u\dot{\sigma}_u + \sigma_\psi\dot{\sigma}_\psi + \sigma_r\dot{\sigma}_r + \zeta_r\dot{\zeta}_r - \frac{\tilde{\rho}_u\dot{\rho}_u}{\lambda_0} - \frac{\tilde{\rho}_r\dot{\rho}_r}{\eta_0} \quad (39)$$

Based on the definitions of ζ_r , its derivative is given by

$$\dot{\zeta}_r = -\frac{1}{\kappa_r} (\text{sig}^{2p-1}(\zeta_r) + \text{sig}^{2q-1}(\zeta_r)) - \dot{r}_c. \quad (40)$$

In the light of (32), we can assume that $\dot{r}_c$ is bounded, i.e. $|\dot{r}_c| < \epsilon_r, \epsilon_r > 0$. Then, one has

$$\dot{\zeta}_r \leq -\frac{1}{\kappa_r} (\text{sig}^{2p-1}(\zeta_r) + \text{sig}^{2q-1}(\zeta_r)) + \epsilon_r. \quad (41)$$

Substituting (25)–(27), (31)–(32), (35)–(37) and (41) into (39) yields

$$\begin{aligned} \dot{V} &= \sigma_u (d_u - \hat{\rho}_u \text{sign}(\sigma_u) - k_{1u} \text{sig}^{2p-1}(\sigma_u) - k_{2u} \text{sig}^{2q-1}(\sigma_u)) \\ &\quad + \sigma_\psi (-l_{1\psi} \text{sig}^{2p-1}(\sigma_\psi) - l_{2\psi} \text{sig}^{2q-1}(z_\psi)) \\ &\quad + \sigma_r (d_r - \hat{\rho}_r \text{sign}(\sigma_r) - k_{1r} \text{sig}^{2p-1}(\sigma_r) - k_{2r} \text{sig}^{2q-1}(\sigma_r)) \\ &\quad + \zeta_r \left(-\frac{1}{\kappa_r} (\text{sig}^{2p-1}(\zeta_r) + \text{sig}^{2q-1}(\zeta_r)) - \dot{r}_c \right) - \frac{\tilde{\rho}_u\dot{\rho}_u}{\lambda_0} - \frac{\tilde{\rho}_r\dot{\rho}_r}{\eta_0} \\ &\leq -k_{1u}\sigma_u^{2p} - k_{2u}\sigma_u^{2q} - l_{1\psi}\sigma_\psi^{2p} - l_{2\psi}\sigma_\psi^{2q} - k_{1r}\sigma_r^{2p} - k_{2r}\sigma_r^{2q} - \frac{1}{\kappa_r}\zeta_r^{2p} \\ &\quad - \frac{1}{\kappa_r}\zeta_r^{2q} + |\zeta_r|\epsilon_r - \frac{\tilde{\rho}_u}{\lambda_0} (\dot{\rho}_u - \lambda_0|\sigma_u|) - \frac{\tilde{\rho}_r}{\eta_0} (\dot{\rho}_r - \eta_0|\sigma_r|) \\ &\leq -k_{1u}\sigma_u^{2p} - k_{2u}\sigma_u^{2q} - l_{1\psi}\sigma_\psi^{2p} - l_{2\psi}\sigma_\psi^{2q} - k_{1r}\sigma_r^{2p} - k_{2r}\sigma_r^{2q} - \frac{1}{\kappa_r}\zeta_r^{2p} \\ &\quad - \frac{1}{\kappa_r}\zeta_r^{2q} + \frac{1}{2}\zeta_r^{2p} + \frac{1}{2}\zeta_r^{2q} + \frac{1}{2}\epsilon_r^2 + \frac{\lambda_1}{\lambda_0}\tilde{\rho}_u\dot{\rho}_u + \frac{\lambda_2}{\lambda_0}\tilde{\rho}_u\dot{\rho}_u^{2q-1} \\ &\quad + \frac{\eta_1}{\eta_0}\tilde{\rho}_r\dot{\rho}_r + \frac{\eta_2}{\eta_0}\tilde{\rho}_r\dot{\rho}_r^{2q-1}. \end{aligned} \quad (42)$$

According to lemma 2.2 in [32], we have

$$\left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^p \leq \frac{\tilde{\rho}_u^2}{2\lambda_0} + (1-p)p^{\frac{p}{1-p}} \quad (43)$$

$$\left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^p \leq \frac{\tilde{\rho}_r^2}{2\eta_0} + (1-p)p^{\frac{p}{1-p}}. \quad (44)$$

According to lemma 3 in [33], one has

$$\tilde{\rho}_u\hat{\rho}_u^{2q-1} = \tilde{\rho}_u(\rho_u - \tilde{\rho}_u)^{2q-1} \leq \frac{2q-1}{2q}(\rho_u^{2q} - \tilde{\rho}_u^{2q}) \quad (45)$$

$$\tilde{\rho}_r\hat{\rho}_r^{2q-1} = \tilde{\rho}_r(\rho_r - \tilde{\rho}_r)^{2q-1} \leq \frac{2q-1}{2q}(\rho_r^{2q} - \tilde{\rho}_r^{2q}). \quad (46)$$

Substituting (43)–(46) into (42) yields

$$\dot{V} \leq -k_{1u}\sigma_u^{2p} - k_{2u}\sigma_u^{2q} - l_{1\psi}\sigma_\psi^{2p} - l_{2\psi}\sigma_\psi^{2q} - k_{1r}\sigma_r^{2p} - k_{2r}\sigma_r^{2q} - \frac{1}{\kappa_r}\zeta_r^{2p} \quad (47)$$

$$\begin{aligned} &- \frac{1}{\kappa_r}\zeta_r^{2q} + \frac{1}{2}\zeta_r^{2p} + \frac{1}{2}\zeta_r^{2q} + \frac{1}{2}\epsilon_r^2 - \lambda_1 \left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^p - \frac{2^q\lambda_0^{q-1}\lambda_2(2q-1)}{2q} \left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^q \\ &+ \frac{\lambda_1\rho_u^2}{2\lambda_0} + \frac{\lambda_2(2q-1)}{2q\lambda_0}\rho_u^{2q} + \lambda_1(1-p)p^{\frac{p}{1-p}} - \eta_1 \left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^p - \frac{2^q\eta_0^{q-1}\eta_2(2q-1)}{2q} \left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^q \\ &+ \frac{\eta_1\rho_r^2}{2\eta_0} + \frac{\eta_2(2q-1)}{2q\eta_0}\rho_r^{2q} + \eta_1(1-p)p^{\frac{p}{1-p}} \\ &\leq -A \left(\left(\frac{1}{2}\sigma_u^2 \right)^p + \left(\frac{1}{2}\sigma_\psi^2 \right)^p + \left(\frac{1}{2}\sigma_r^2 \right)^p + \left(\frac{1}{2}\zeta_r^2 \right)^p + \left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^p + \left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^p \right) \\ &- B \left(\left(\frac{1}{2}\sigma_u^2 \right)^q + \left(\frac{1}{2}\sigma_\psi^2 \right)^q + \left(\frac{1}{2}\sigma_r^2 \right)^q + \left(\frac{1}{2}\zeta_r^2 \right)^q + \left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^q + \left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^q \right) + \Delta \end{aligned} \quad (48)$$

$$- B \left(\left(\frac{1}{2}\sigma_u^2 \right)^q + \left(\frac{1}{2}\sigma_\psi^2 \right)^q + \left(\frac{1}{2}\sigma_r^2 \right)^q + \left(\frac{1}{2}\zeta_r^2 \right)^q + \left(\frac{\tilde{\rho}_u^2}{2\lambda_0} \right)^q + \left(\frac{\tilde{\rho}_r^2}{2\eta_0} \right)^q \right) + \Delta \quad (49)$$

where

$$\begin{aligned}
 A &= \min \left\{ 2^p k_{1u}, 2^p l_{1\psi}, 2^p k_{1r}, 2^p \left(\frac{1}{\kappa_r} - \frac{1}{2} \right), \lambda_1, \eta_1 \right\} \\
 \bar{B} &= \min \left\{ 2^q k_{2u}, 2^q l_{2\psi}, 2^q k_{2r}, 2^q \left(\frac{1}{\kappa_r} - \frac{1}{2} \right), \right. \\
 &\quad \left. \frac{2^q \lambda_0^{q-1} \lambda_2 (2q-1)}{2q}, \frac{2^q \eta_0^{q-1} \eta_2 (2q-1)}{2q} \right\} \\
 \Delta &= \frac{1}{2} \epsilon_r^2 + \frac{\lambda_1 \rho_u^2}{2\lambda_0} + \frac{\lambda_2 (2q-1)}{2q\lambda_0} \rho_u^{2q} + \lambda_1 (1-p) p^{\frac{p}{1-p}} \\
 &\quad + \frac{\eta_1 \rho_r^2}{2\eta_0} + \frac{\eta_2 (2q-1)}{2q\eta_0} \rho_r^{2q} + \eta_1 (1-p) p^{\frac{p}{1-p}}. \quad (50)
 \end{aligned}$$

According to lemma 1 in [31], one has

$$\begin{aligned}
 \dot{V} &\leq -A \left(\frac{1}{2} \sigma_u^2 + \frac{1}{2} \sigma_\psi^2 + \frac{1}{2} \sigma_r^2 + \frac{1}{2} \zeta_r^2 + \frac{\tilde{\rho}_u^2}{2\lambda_0} + \frac{\tilde{\rho}_r^2}{2\eta_0} \right)^p \\
 &\quad - 6^{1-q} \bar{B} \left(\frac{1}{2} \sigma_u^2 + \frac{1}{2} \sigma_\psi^2 + \frac{1}{2} \sigma_r^2 + \frac{1}{2} \zeta_r^2 + \frac{\tilde{\rho}_u^2}{2\lambda_0} + \frac{\tilde{\rho}_r^2}{2\eta_0} \right)^q + \Delta \\
 &= -AV^p - BV^q + \Delta \quad (51)
 \end{aligned}$$

where $B = 6^{1-q} \bar{B}$.

According to Lemma 1 in [34], system (38) is practical fixed time stable and will converge to the neighborhood of zero within in a fixed-time, awchich given by

$$t_s \leq T_{\max} = \frac{1}{A\omega(1-p)} + \frac{1}{B\omega(q-1)} \quad (52)$$

where $0 < \omega < 1$ is a constant. This completes the proof. $\square$

5. Illustrative simulation

In this section, the effectiveness of the proposed intelligent navigation approach will be confirmed by six simulations. The initial three simulations aim to confirm the effectiveness of the improved APF algorithm in three scenarios: target unreachable, local minimum, and dynamic obstacles avoidance. The final three simulations consist of the waypoints tracking under the obstacle-free, the static obstacles, and the dynamic obstacles environment. The parameters of the improved APF algorithm, the AUV model and the AFISM scheme are given in table 1.

5.1. Path planning algorithm comparison simulations

Scenario 1: GNRON problem

In this scenario, the AUV initial position is set as (0 m, 0 m), the AUV initial heading angle is set as 0 rad, the obstacle centre position is set as (115 m, 118 m), the obstacle radius is set as 30 m, and the target position is set as (140 m, 140 m). As shown in figure 14(a), when the obstacle is close to the target, under the action of traditional APF algorithm, as the target position is very close to the obstacle, the repulsive force generated by the obstacle is greater than the attractive force generated by the target when the AUV moves

towards the target point, resulting in the AUV oscillating around the target point. However, under the improved APF algorithm, the AUV oscillates around the target point. The repulsive and attractive forces are adjusted by introducing logarithmic functions and distance equilibrium terms. Therefore, the AUV can reach the target point smoothly, as shown in figure 14(b).

Scenario 2: Local minimum

In this scenario, the AUV initial position is set as (0 m, 0 m), the AUV initial heading angle is set as $\pi/4$ rad, and target position is (140 m, 140 m). In a case of a single obstacle, the obstacle centre position is set as (75 m, 75 m), the obstacle radius is set as 22m. In the case of multiple obstacles, the obstacle centre position is set as (55 m, 95 m) and (95 m, 55 m), the obstacle radius is set as 18m.

From figures 15(a) and 16(a), it can be seen that when the obstacle is in a special position, the attractive and repulsive forces on the AUV cancel each other out. The traditional APF algorithm is stuck when the potential field has a local minimum value, but by constructing a virtual barrier circle, the situation that the attraction and repulsion cancel out is broken, so that the AUV can pass through the local minimum region, as shown in figures 15(b) and 16(b).

Scenario 3: Dynamic obstacles avoidance

In this scenario, we design two simulations of AUV approaching and moving away from dynamic obstacles. the initial position of the AUV is set as (0 m, 0 m), the initial heading angle of the AUV is set as 0 rad, the initial position of the obstacle is set as (105 m, 52 m) and (12 m, 52 m), the radius of the obstacle is set as 8 m, the speed of the obstacle is set as -5 m s^{-1} and 3 m s^{-1} , the start time of the movement of the obstacle is set as 15.5 s, and the target position is set as (100 m, 100 m).

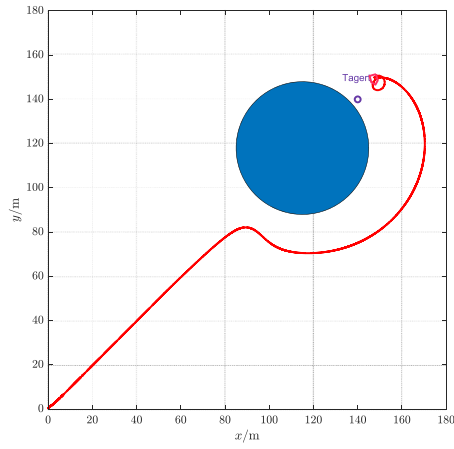
The traditional APF is not suitable for dynamic obstacles avoidance as it does not consider the motion states of the AUV and obstacles. This can result in collisions when the obstacle is close enough to require an urgent change of course, as illustrated in figure 17. The improved APF algorithm determines the need to avoid dynamic obstacles by comparing the angle between the relative velocity direction and the relative position direction, and the angle between the relative position and the tangent line of the safety range. This allows for a safe and smooth avoidance of dynamic obstacles by the AUV, as demonstrated in figure 18.

When the AUV is far away from the obstacle, the traditional APF algorithm does not consider the direction and speed of the obstacle, but only considers the position relationship between the AUV and the obstacle. This causes the AUV to move in the same direction as the obstacle, greatly increasing the AUV's final path length, as illustrated in figure 19. Under the influence of the velocity potential function, the improved APF enables the AUV to find a shorter and safer path, as illustrated in figure 20.

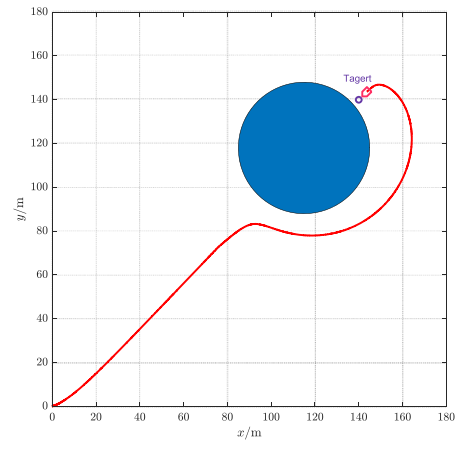
The above simulations verify that the improved APF algorithm proposed in this paper overcomes the limitations of the traditional APF algorithm. For GNRON problem, by introducing logarithmic function and distance compensation

Table 1. table of simulation parameters.

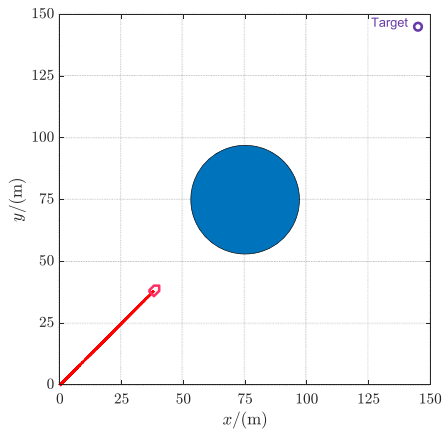
Indexes	Items
Improved APF parameters	$k_{att1} = 0.15, k_{att2} = 0.1, k_{rep1} = 162, k_{rep2} = 60$ $d_t = 20, k_v = 100, d_0 = 2.5R, O_i = 5$
AUV model parameters	$m_{11} = 25.8 \text{ kg}, m_{22} = 2.76 \text{ kg}, d_{11} = 12 \text{ kgs}^{-1}, d_{uu} = 2.5 \text{ kgs}^{-1}$ $d_{22} = 17 \text{ kgs}^{-1}, d_{vv} = 4.5 \text{ kgs}^{-1}, d_{33} = 0.5 \text{ kgs}^{-1}, d_{rr} = 0.1 \text{ kgs}^{-1}$
AFISMC parameters	$p = 0.6, q = 1.5, m = 0.5, n = 1.2, k_{1u} = 1, k_{2u} = 1$ $\lambda_0 = 2, \lambda_1 = 0.02, \lambda_2 = 0.02, \eta_0 = 0.5, \eta_1 = 0.01, \eta_2 = 0.01$ $a_u = 0.5, b_{1u} = 1, b_{2u} = 1, \alpha_\psi = 0.1, b_{1\psi} = 1, b_{2\psi} = 1$ $l_{1\psi} = 0.5, l_{2\psi} = 0.5, k_{1r} = 0.1, k_{2r} = 0.1$



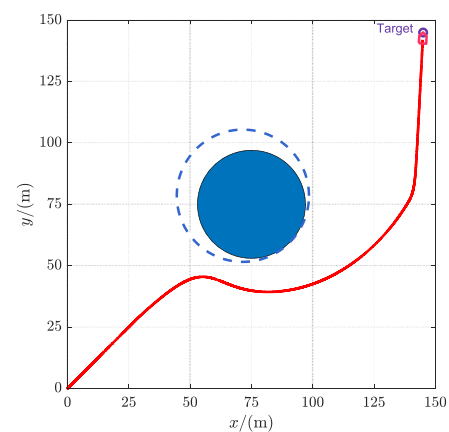
(a) Traditional APF algorithm



(b) Improved APF algorithm

Figure 14. Simulation results of the GNRON problem.

(a) Traditional APF algorithm



(b) Improved APF algorithm

Figure 15. Simulation results of the local minimum problem under single obstacle.

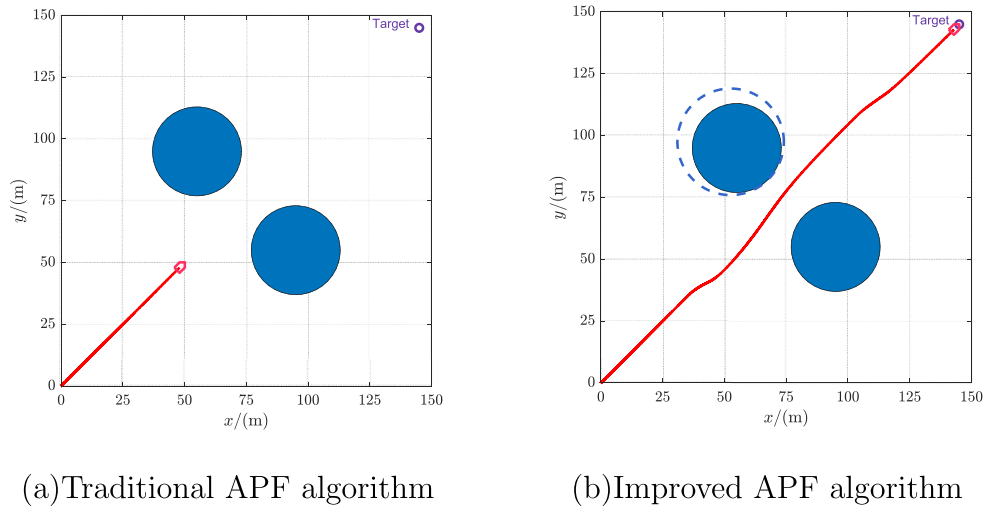


Figure 16. Simulation results of the local minimum problem under multiple obstacles.

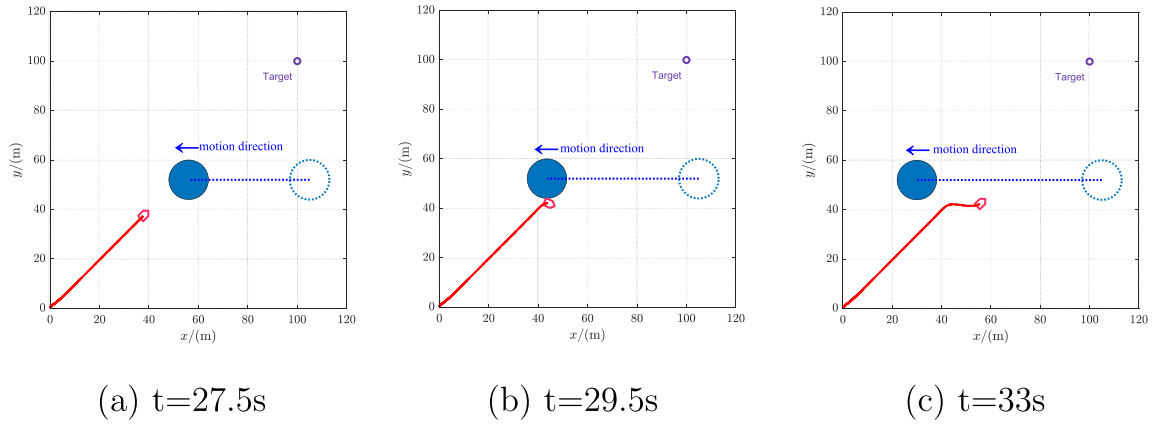


Figure 17. Simulation results of dynamic obstacle avoidance under traditional APF algorithm (AUV approaching obstacles).

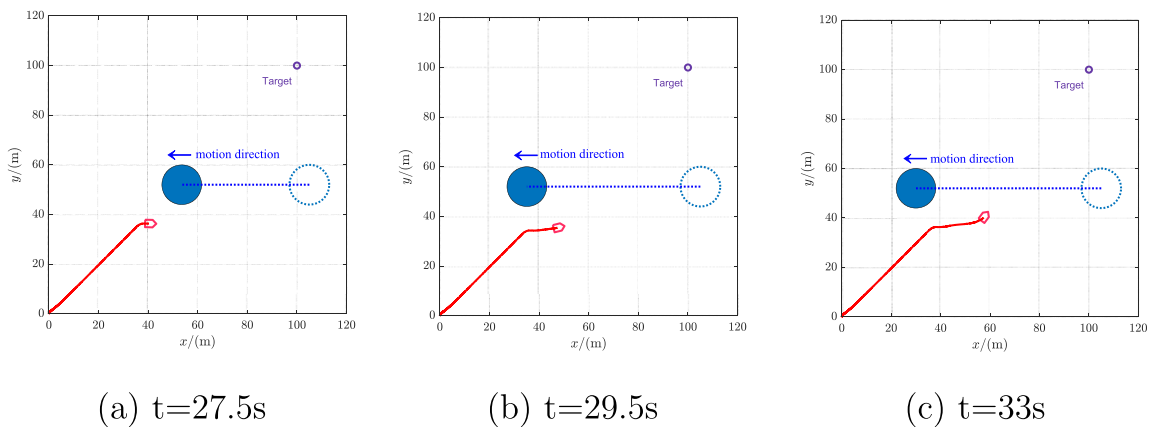


Figure 18. Simulation results of dynamic obstacle avoidance under improved APF algorithm (AUV approaching obstacles).

term, the AUV can reach the target when the obstacle is near the target. The proposed virtual obstacle construction method ensures that the AUV is not fall into the local minimum.

Furthermore, the influence of the velocities of the AUV and obstacle is considered and the velocity repulsive potential field is added to solve the dynamic obstacles avoidance problem.

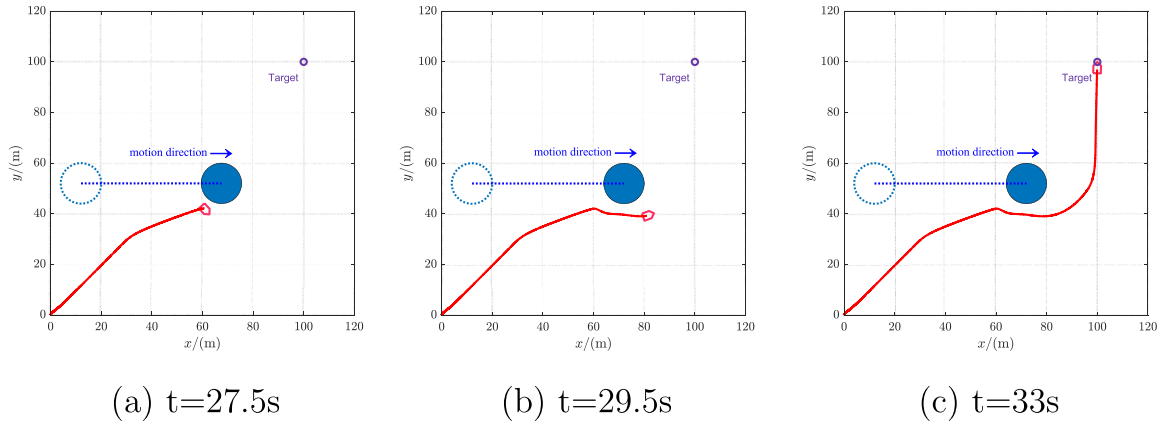


Figure 19. Simulation results of dynamic obstacle avoidance under traditional APF algorithm (AUV deviation obstacle).

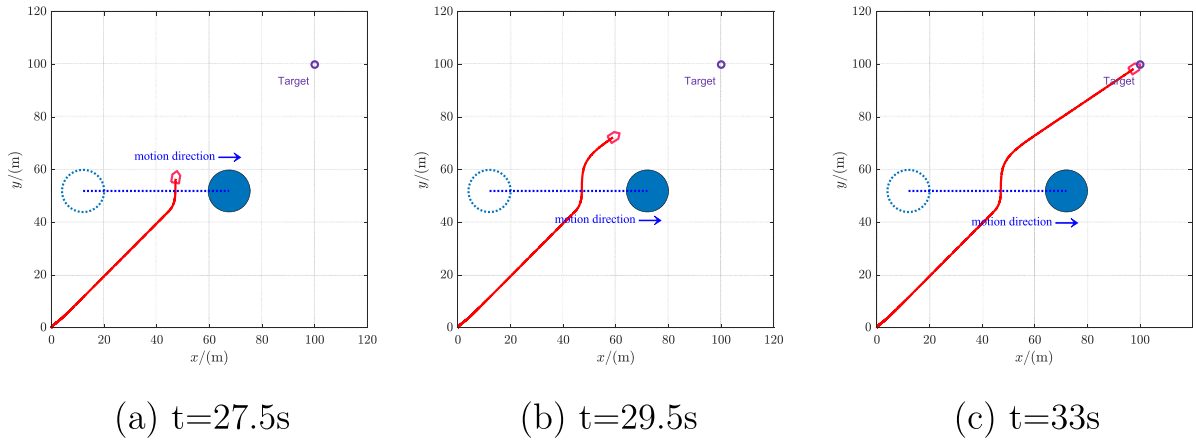


Figure 20. Simulation results of dynamic obstacle avoidance under improved APF algorithm (AUV deviation obstacle).

5.2. Navigational simulations

This section presents three simulations designed to verify the availability and practicality of the proposed integrated real-time path planning and tracking control strategy in complex environments. The simulation is conducted in a hypothetical ocean with external disturbances. The external disturbance is set according to [21]:

$$\begin{bmatrix} d_u \\ d_v \\ d_r \end{bmatrix} = \begin{bmatrix} 0.5 + 0.2 \sin(0.02t) + 0.2 \sin(0.1t)N \\ -0.3 + 0.2 \sin(0.02t - \frac{\pi}{6}) + 0.15 \sin(0.3t)N \\ -0.4 - 0.2 \sin(0.05t) - 0.2 \sin(0.1t)N \cdot m \end{bmatrix}. \quad (53)$$

Scenario 1: waypoints tracking in obstacle-free environment

For verifying the performance of the proposed controller in the presence of unknown disturbances, obstacle-free environment is considered in the first simulation. The initial position and initial heading angle of the AUV are set as (5 m, 5 m) and 0 rad, and three target path points are set as (50 m, 150 m), (200 m, 200 m), (295 m, 295 m). The simulation results are shown in the figures 21 and 22.

Scenario 2: Waypoints tracking in static obstacles environment

This scenario considers the waypoints tracking under static obstacle environment to verify the performance of the proposed real-time path planning and control scheme. In this simulation, a large number of static obstacles are set up. The initial status of the AUV and setting of waypoints is the same as those in Scenario 1. The simulation results are shown in the figures 23 and 24.

Scenario 3: waypoints tracking in dynamic obstacles environment

To further examine the capability of the proposed method, this scenario considers the problem of tracking waypoints in a dynamic obstacle environment. The simulation results are shown in the figures 25 and 26.

The simulation results for the three scenarios are shown in figures 21, 23 and 25, which show that the proposed integrated real-time path-planning and control framework based on the improved APF algorithm can guide the AUV through the obstacle region to reach each target waypoint in various complex obstacle environments. The tracking errors $\tilde{u}$, $\tilde{v}$ and control inputs τ_u , τ_r are given in figures 22, 24 and 26. As

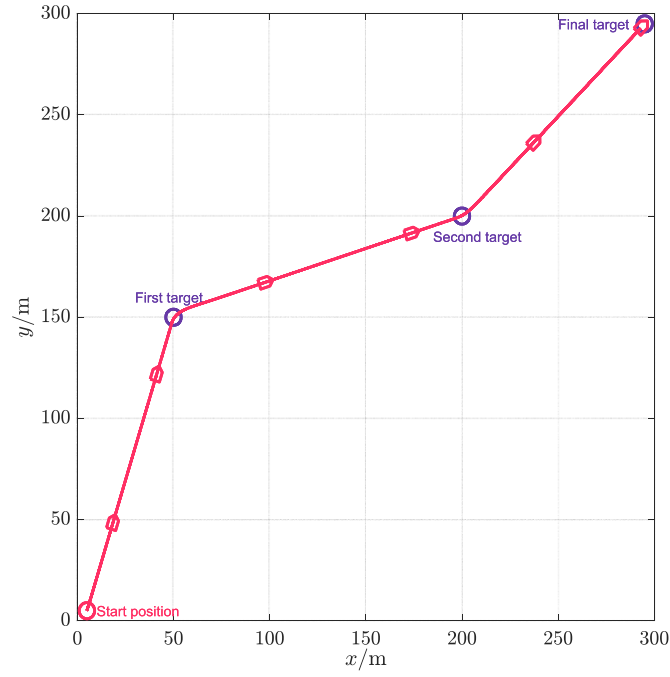


Figure 21. Waypoints tracking in obstacle-free environment in scenario 1.

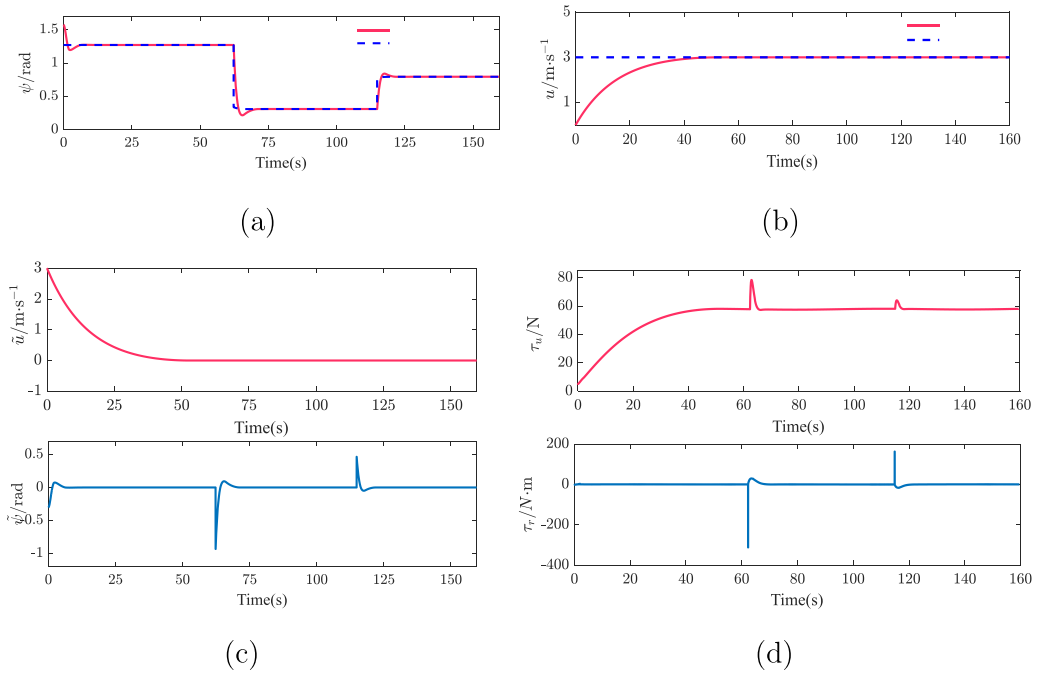


Figure 22. Simulation results of waypoints tracking in obstacle-free environment in scenario 1. (a) Actual and desired heading angle. (b) Actual and desired surge speed. (c) Tracking errors $\dot{u}$ and $\dot{\psi}$. (d) Control force τ_u and moment τ_r .

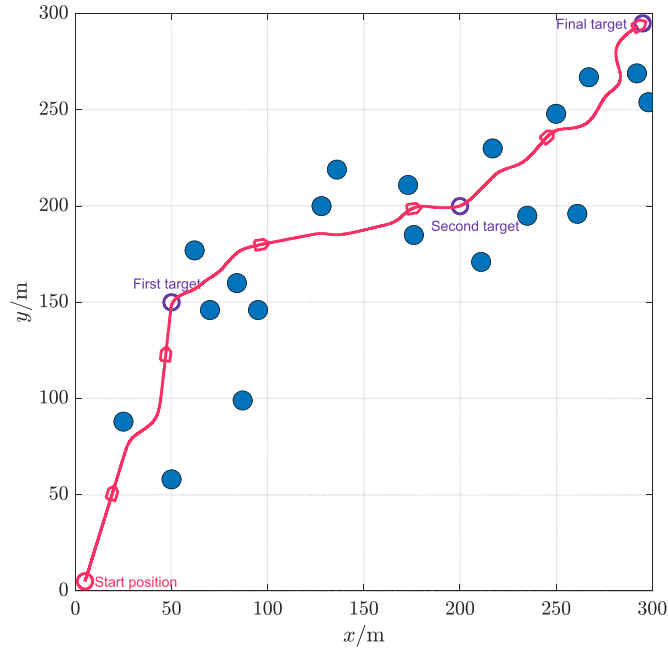


Figure 23. Waypoints tracking in static obstacles environment in scenario 2.

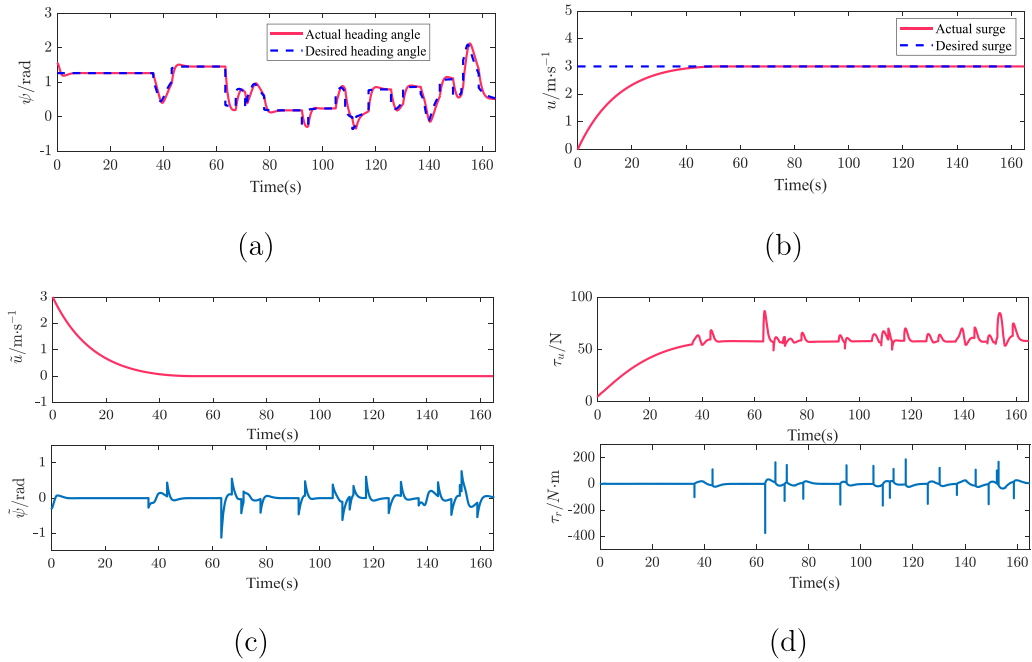


Figure 24. Simulation results of waypoints tracking in static obstacles environment in scenario 2. (a) Actual and desired heading angle. (b) Actual and desired surge speed. (c) Tracking errors $\tilde{u}$ and $\tilde{\psi}$ (d) control force τ_u and moment τ_r .

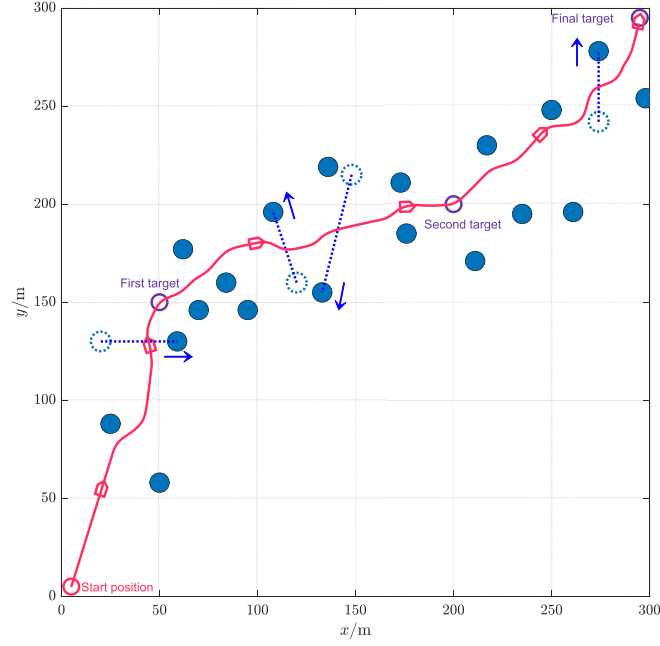


Figure 25. Waypoints tracking in dynamic obstacles environment in scenario 3.

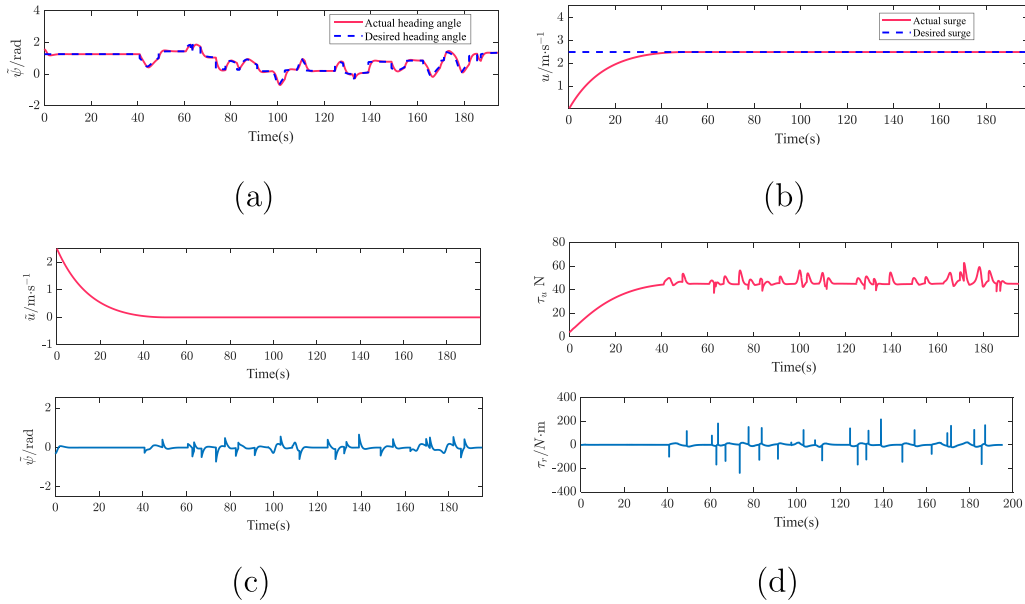


Figure 26. Simulation results of waypoints tracking in dynamic obstacles environment in scenario 3. (a) Actual and desired heading angle. (b) Actual and desired surge speed. (c) Tracking errors $\tilde{u}$ and $\tilde{\psi}$ (d) control force τ_u and moment τ_r .

shown in figure 22, the proposed AFISM can accurately track the desired heading and speed of the AUV when it is subjected to external disturbances, while keeping the control inputs within an acceptable saturation range. In figures 24 and 26, it can be seen that the heading angle changes more frequently with the increase of obstacles, so the above control algorithm produces heading angle errors and control input fluctuations. But they are within acceptable limits.

6. Conclusion

In this paper, an integrated real-time path planning and tracking framework based on an improved APF algorithm is proposed for AUVs in complex ocean environments. An improved APF algorithm is proposed to achieve real-time path planning for the AUV. Compared with the traditional APF algorithm, the proposed improved APF algorithm overcomes the problem

of GNRON, local minimum and dynamic obstacles avoidance by introducing and combining a logarithmic function, a distance compensation term, a virtual obstacle construction method, and a velocity repulsive potential field function. Moreover, an AFISM scheme is presented for tracking the planned path of collision avoidance manoeuvres. Stability analysis proves that the tracking errors of the AUV are stabilized towards the small neighbourhood of the origin in a fixed time independent of the initial conditions. Simulation results illustrate that the proposed real-time path planning and tracking control strategy can make the AUV track the desired collision-free path.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 52101401 and No. 52176077) and The Youth Innovation Project for Universities of Shandong Province (2024KJH102).

ORCID iD

Shun An  <https://orcid.org/0000-0003-0584-0554>

References

- [1] Wu B, Han X, Hui N 2021 System identification and controller design of a novel autonomous underwater vehicle *Machines* **9** 109
- [2] Li J, Fu J, Zou N, Qiu L and Zhang X 2024 High frame rate acoustic navigation of AUV based on range ambiguity suppression *Meas. Sci. Technol.* **35** 076303
- [3] Agrawal P and Agrawal H 2018 Adaptive algorithm design for cooperative hunting in multi-robots *Int. J. Intell. Control Syst.* **10** 12
- [4] Huang Z, Zhu D and Sun B 2016 A multi-AUV cooperative hunting method in 3-D underwater environment with obstacle *Eng. Appl. Artif. Intell.* **50** 192–200
- [5] Liang X, Liu Y-H, Wang H, Chen W, Xing K and Liu T 2016 Leader-following formation tracking control of mobile robots without direct position measurements *IEEE Trans. Autom.* **61** 4131–7
- [6] Zhao Z, Hu Q, Feng H, Feng X and Su W 2022 A cooperative hunting method for multi-AUV swarmin underwater weak information environment with obstacles *J. Mar. Sci. Eng.* **10** 1266
- [7] Cheng C, Sha Q, He B and Li G 2021 Path planning and obstacle avoidance for AUV: a review *Ocean Eng.* **235** 109355
- [8] Teng M, Ye L, Yanqing J, Rupeng W, Zheng C and Yusen G 2018 A dynamic path planning method for terrain-aided navigation of autonomous underwater vehicles *Meas. Sci. Technol.* **29** 095105
- [9] Yang L, Yang M and Wang A 2024 A multi-robot conflict elimination path planning approach for dynamic environments *Meas. Sci. Technol.* **36** 016340
- [10] Taheri E, Ferdowsi M H and Danesh M 2019 Closed-loop randomized kinodynamic path planning for an autonomous underwater vehicle *Appl. Ocean Res.* **83** 48–64
- [11] Ni J, Wu L, Shi P and Yang S X 2017 A dynamic bioinspired neural network based real-time path planning method for autonomous underwater vehicles *Comput. Intell. Neurosci.* **2017** 9269742
- [12] Khatib O 1986 Real-time obstacle avoidance for manipulators and mobile robots *Int. J. Rob. Res.* **5** 90–98
- [13] Viturino C C B, Junior U A G S and Schnitman L 2020 Adaptive artificial potential fields with orientation control applied to robotic manipulators *IFAC-PapersOnLine* **53** 9924–9
- [14] Zhang W, Wang N and Wu W 2023 A hybrid path planning algorithm considering AUV dynamic constraints based on improved A* algorithm and APF algorithm *Ocean Eng.* **285** 115333
- [15] Zhang W, Xu G, Song Y and Wang Y 2023 An obstacle avoidance strategy for complex obstacles based on artificial potential field method *J. Field Robot.* **40** 1231–44
- [16] He Z, Chu X, Liu C and Wu W 2023 A novel model predictive artificial potential field based ship motion planning method considering colregs for complex encounter scenarios *ISA Trans.* **134** 58–73
- [17] Yu H and Ning L 2023 Coordinated obstacle avoidance of multi-AUV based on improved artificial potential field method and consistency protocol *J. Mar. Sci. Eng.* **11** 1157
- [18] Zhang H and Luo F 2022 An improved UAV path planning method based on APSOvnp-APF algorithm (IEEE) pp 5458–63
- [19] Luo J, Wang Z-X and Pan K-L 2022 Reliable path planning algorithm based on improved artificial potential field method *IEEE Access* **10** 108276–84
- [20] Li B, Gao X, Huang H and Yang H 2024a Improved adaptive twisting slidingmode control for trajectory tracking of an AUV subject to uncertainties *Ocean Eng.* **297** 116204
- [21] Zhang X and Jiang K 2024 Backstepping-based adaptive control of underactuated AUV subject to unknown dynamics and zero tracking errors *Ocean Eng.* **302** 117640
- [22] Liu X, Sun B and Su Z 2024 Adaptive sliding mode control for AUV based on backstepping and neural networks *Meas. Sci. Technol.* **36** 016215
- [23] Zhang Y, Liu J, Yu J and Liu D 2023c Single neural network-based asymptotic adaptive control for an autonomous underwater vehicle with uncertain dynamics *Ocean Eng.* **286** 115553
- [24] Thanh P N N and Anh H P H 2023 Neural network-based depth and horizontal control for autonomous underwater vehicles with prescribed performance *Ocean Eng.* **281** 114647
- [25] Pan X, Xu G, Huang Z and Xie S 2022 Adaptive fuzzy control design based on nonlinear system of underwater vehicle *Isope (ISOPE-I)*
- [26] Elmokadem T, Zribi M and Youcef-Toumi K 2016 Trajectory tracking sliding mode control of underactuated AUVs *Nonlinear Dyn.* **84** 1079–91
- [27] Luo W and Liu S 2023 Disturbance observer based nonsingular fast terminal sliding mode control of underactuated AUV *Ocean Eng.* **279** 114553
- [28] Esfahani Z F, Vali A R, Hakimi S M, BehnamGol V and Abdi B 2023 Slidingmode controller design based on state

- estimation for underactuated AUVs in the presence of time-varying disturbances and uncertainties *Int. J. Dyn. Control* **11** 1637–52
- [29] Wang Q, Wu Z, Xie M, Wu F and Huang H 2023 Finite-time prescribed performance trajectory tracking control for the autonomous underwater helicopter *Ocean Eng.* **280** 114628
- [30] Fossen T I 2011 Handbook of marine craft hydrodynamics and motion control
- [31] Zuo Z 2015 Nonsingular fixed-time consensus tracking for second-order multi-agent networks *Automatica* **54** 305–9
- [32] Xu B, Liang Y, Li Y X and Hou Z 2022 Adaptive command filtered fixed-time control of nonlinear systems with input quantization *Appl. Math. Comput.* **427** 127186
- [33] Wang F and Lai G 2020 Fixed-time control design for nonlinear uncertain systems via adaptive method *Syst. Control Lett.* **140** 104704
- [34] Kim H J and Yoo S J 2024 Adaptive fixed-time containment control of uncertain underactuated underwater vehicles under dynamic event-driven mechanism *Ocean Eng.* **291** 116488